

Qatar-1b

R-0606

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230526 · created 2026-07-25T06:08:17+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460090.8866 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.056
Transit depth (Rp/Rs)^2	3.61 +/- 2.13 [%]
Orbital Inclination (inc)	79.74 +/- 2.82
Airmass coefficient 1 (a1)	5.56 +/- 3.52
Airmass coefficient 2 (a2)	-1.33 +/- 0.94
Scatter in the residuals of the lightcurve fit is	63.99 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	7.5
Transit Duration (day)	0.031 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.056 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.031 d vs 0.0692 d (deviation 1.32σ)
Mid-time O–C	-4.2 min
Depth SNR	3.4
Residual scatter	63.99 %

Light curve

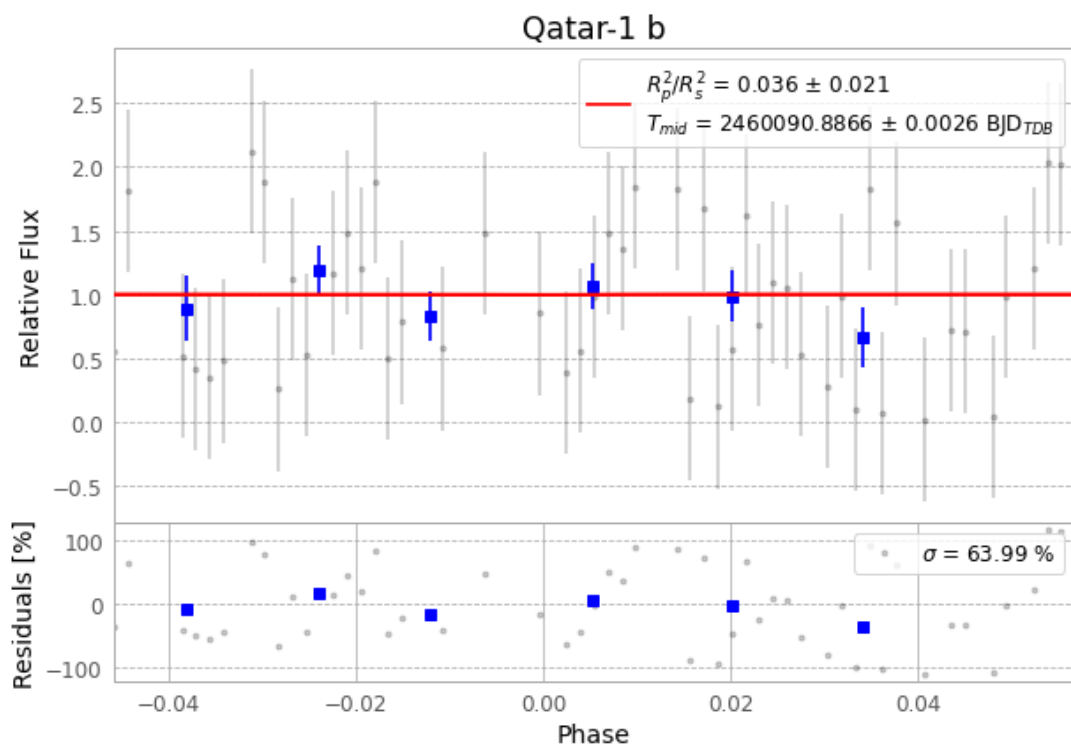


Figure 1 — FinalLightCurve_Qatar-1 b_2023-05-26.png (SHA-256 e09b1363fe29e89e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [248, 299], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8868c6a5b01a46d4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

74 FITS frames (49 MB), Qatar-1230526073315.FITS → Qatar-1230526111215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230526.log (a8bd6f110535...), FinalLightCurve_Qatar-1 b_2023-05-26.png (e09b1363fe29...), FinalLightCurve_Qatar-1 b_2023-05-26.csv (fcae2baf4dd2...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 06:08Z.